

# Md Omar Faruqe Riyad

Software Engineer | Backend Systems & AI Infrastructure

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## EDUCATION

**Shahjalal University of Science and Technology (SUST)**

Sylhet, Bangladesh

*B.Sc. in Computer Science and Engineering; CGPA: 3.53/4.00 (Last two years: 3.83)*

2018 – 2023

## EXPERIENCE

**Dynamic Solution Innovators** 

April 2023 – Present

*Software Engineer*

*Dhaka, Bangladesh*

- **EC-BVRS:** Contributed to the data pipeline behind the voter registration flow of Bangladesh's national voter registration (NID) platform, which processes records for nearly **150 million voters** (Spring Boot, Kafka, Redis).
- **Tooling:** Built and open-sourced [gf](#) , a GlassFish CLI that hot-swaps modified bytecode into running JVMs over JDI/JDWP and bypasses JasperReports classloader caching. Cut local edit-deploy cycles from **~2 minutes** to **~5 seconds**.
- **E-Appeal (NBR):** Consolidated authorization checks scattered across JSF views and service code into a single permission-evaluation layer for role-based access control.
- **SHMS:** Built a configurable approval-workflow engine that replaced hardcoded per-module approval logic. Added a field-level correction flow so reviewers can reopen individual form fields with inline feedback instead of rejecting whole submissions.
- **SHMS:** Introduced Prometheus, Loki, and Grafana observability for GlassFish applications, exposing JVM memory, thread pools, and request latency for production debugging and capacity planning.
- **BCIC-ERP:** Replaced base64-inlined file rendering with URL-based streaming behind a pluggable **StorageClient** interface (MinIO or filesystem), cutting page payload size and server memory use. Files are served through nginx X-Accel-Redirect after per-request authorization checks.
- **BCIC-ERP:** Moved SMS and email notifications onto a JMS queue with dead-letter retries, so provider outages no longer block user-facing requests. Led development of the Procurement, Inventory, Asset, and Budget modules (4 of 12), from process modeling and schema design through production rollout.
- **Cross-project:** Isolated auditing and notification dispatch from business logic with AspectJ cross-cutting interceptors, a pattern reused in multiple Jakarta EE systems.

**Altri.ai** 

Part-time

*Full-Stack AI Engineer, RevestAI* 

*Remote, US*

- Rewrote the sequential LLM enrichment pipeline as bounded-concurrency async batches with atomic per-batch commits. Throughput rose about **10×**, and failed runs resume without repeating completed LLM calls.
- Designed the three-stage property scoring pipeline that separates LLM feature extraction, ML valuation, and deterministic financial heuristics, so each layer can be retrained and tested on its own.
- Built the property ingestion pipeline on HomeHarvest and PostgreSQL with content-hash deduplication and incremental delta-syncing. Repeated scrapes no longer trigger duplicate enrichment runs, and historical listing state is preserved.
- Implemented distributed job coordination on PostgreSQL with **SELECT FOR UPDATE** row locks and stale-lock recovery, so multiple scheduler replicas safely share long-running enrichment and training jobs.
- Built the training and promotion pipeline for XGBoost property valuation. A challenger must beat the champion on a gold-standard holdout slice, and a human signs off before production-wide rescoring, since aggregate metrics can hide slice regressions and valuation errors cost money.

## PROJECTS

**SREGym Fault Scenario Contribution**  | *Python, Kubernetes, Docker*

2026

- Reproduced a documented Kubernetes admission-control incident ([issue #128162](#)) where cumulative mutating-webhook timeouts exceed the API server's admission deadline and silently block all pod creation. Merged into SREGym, an open-source benchmark for AI SRE agents.
- Designed a four-property mitigation oracle with a fresh probe-pod admission test that rejects reward-hacking fixes, such as deleting all webhooks, that mask the symptom without restoring the control-plane network path.

**Token-Efficient LLM Query Router**  | *AMD Hackathon Act II, Track 1*

2026

- Built a precision-gated routing cascade that answers 8 task categories while minimizing paid API tokens: a zero-token regex router (64/64 on dev and adversarial paraphrases), an ONNX sentiment classifier gated to measured 100% precision, a LoRA fine-tuned Qwen2.5-1.5B for local inference, and a cloud model as accuracy backstop. Runtime speed calibration enables a local tier only when the host can finish in time.
- Guarded local answers with deterministic checks instead of model self-assessment, and shipped each submission as an immutable Docker tag. An offline LLM-as-judge harness mirroring the official rubric drove 17 measured releases, including root-cause fixes for a production timeout, an accuracy-gate failure, and a token regression from an over-broad regex.

**KnowledgeRelay** [🔗](#) | *Python, FastAPI, LangChain, ChromaDB, React* 2025

- Built a knowledge-transfer agent that interviews outgoing team members with adaptive questions, ingests their documents and code, and answers new joiners' questions with source attribution. Retrieval rewrites follow-ups into self-contained queries; chunking differs for code and prose.

**PrimeFaces Open Source Contribution** [🔗](#) | *Java, JSF, PrimeFaces* 2024

- Contributed a feature enhancement to PrimeFaces' `JPALazyDataModel` that lets developers inject custom filters into DataTable-backed JPA queries.

## RESEARCH & PUBLICATIONS

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**AI for Site Reliability Engineering (SREGym)** | *Research Intern, advised by Prof. Tianyin Xu* UIUC, 2026

- Working on the SREGym v2 paper: extending the benchmark from synthetic faults to real-world failure scenarios and analyzing how agents behave on them.

**AdversaryAI at BLP-2025 Task 2** [🔗](#) BLP 2025, IJCNLP-AAACL

- Improved low-resource Bangla code generation with LoRA fine-tuning and an execution-guided self-refinement loop (Pass@1 0.85).

**Deja Vu at SemEval 2024 Task 9** [🔗](#) SemEval 2024, NAACL

- Designed data augmentation strategies and compared fine-tuned and prompted language models on lateral-thinking puzzles.

**Team\_Syrax at BLP-2023 Task 1** [🔗](#) BLP 2023, EMNLP

- Built a semi-supervised pipeline with self-training, back-translation, and ensembling for Bangla text classification.

**Visual Question Answering (VQA) Data Synthesis** | *Undergraduate Thesis* 2022

- Generated diverse, context-aware question-answer pairs to widen answer-space coverage in VQA training data.

## HONORS, AWARDS & CERTIFICATIONS

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**4th place** out of 32 teams, Code Generation in Bangla Shared Task, BLP at IJCNLP-AAACL 2025 [\[Leaderboard\]](#)

**6th place** out of 30 teams, DSI AI Agent Hackathon 2025 [\[Leaderboard\]](#)

**12th place** in post-evaluation (19th official), Violence Inciting Text Detection, BLP at EMNLP 2023 [\[Leaderboard\]](#)

**11th place**, BUET CSE Fest 2022 AI Contest

**Certified in Cybersecurity (CC)**, ISC2 [\[Badge\]](#)

## TEACHING

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**Course Instructor** 2024 – Present

*Dynamic Learning* Remote

- Authored and taught [Enterprise Web Development with JEE, JSF, and PrimeFaces](#): component-driven design, CDI/EJB integration, and JSF lifecycle management. [📺](#)

## TECHNICAL SKILLS

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**Languages:** Java, Python, SQL, Bash, JavaScript, C/C++

**Backend & Systems:** Jakarta EE (JSF, CDI, EJB, JPA, JMS), Spring Boot, FastAPI, Kafka, GlassFish, AspectJ

**AI & ML Systems:** PyTorch, LLMs, RAG, LangChain, ChromaDB, XGBoost, LoRA fine-tuning

**Databases & Storage:** PostgreSQL, MySQL, Redis, Supabase, MinIO, Hibernate/JPA, Flyway, Alembic

**Security:** OWASP Top 10, RBAC, Jakarta Security, JWT

**Infrastructure & Observability:** Docker, Kubernetes, Git, GitHub Actions, Prometheus, Grafana, Loki, Nginx, Ansible

**Frontend:** React, PrimeFaces